

Drumming Styles: A Distributional Feature Analysis

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Abstract

This study analyzes the Groove MIDI Dataset from “Learning to Groove with Inverse Sequence Transformations” (Gillick et al., 2019), containing 1,150 MIDI recordings from 10 professional drummers. Principal Component Analysis and Kernel Density Estimation (KDE) is applied to the 446,312 drum hits in the dataset to quantify and identify the latent dimensions and distributional characteristics of the different drummers. Specifically, this study examines velocity dynamics, timing patterns, and drum voice preferences across the 10 drummers and compares the distributional signatures in these features with the identified latent dimensions of playing style. Results reveal that the first three principal components capture 63.8% of variance in drummer features, with PC2 strongly correlating with mean velocity ($r=0.89$). KDE identifies three primary velocity distribution typologies in drummers, suggesting discrete dynamic levels rather than continuous variation. Additionally, this study extends to an application of Hidden Markov Models (HMM) fitted to representative rock recordings for each drummer, which revealed that temporal organization and state-space switching in velocity dynamic levels align with the distributional characteristics. Moreover, the HMM results proved to be a valuable source of nuance that compliments the KDE-derived typologies. Overall, these findings provide interpretable, quantitative characterizations of drumming style that support further research into individual playing styles, particularly across genres.

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Introduction

The motivation behind this study is that we often assume certain aspects of music and certain qualities of drummers. For example, jazz drummers are considered to have a lighter touch and a keen sense of timing and feel. Metal drummers are considered to have speed, endurance, and fast feet. Funk drummers have good command of “the pocket”, meaning that they play tight rhythms that modulate around the hi-hat, snare drum, and bass drum. Overall, this study aims to identify statistically what the true drumming styles are and compare that to traditionally established drumming styles.

Specifically, this study aims to: i) create quantifiable metrics for drumming style, ii) identify latent dimensions of drumming to determine whether differences between drummers are primarily about drum voice preferences, dynamics (velocity), or timing, and iii) determine if drummers have distinct distributional “signatures” in patterns of dynamics (velocity) and timing.

The dataset used is the Groove MIDI (musical instrument data interface) Dataset from “Learning to Groove with Inverse Sequence Transformations” (Gillick et al., 2019). The dataset contains 1,150 MIDI recordings from 10 professional drummers, yielding 446,312 individual drum hits after preprocessing. The MIDI data is event-based data containing individual drum hit information which includes velocity, timing, and pitch. Specifically, the velocity metric can be viewed as how hard the drum was hit and the timing information gives when that hit occurred in the recording. Both are crucial for judging rhythm and dynamics. Additional data includes drummer ID, musical genre assignments (jazz, rock, funk, etc.) and tempo and time signature metadata. A key feature of this dataset is that these recordings are of human drummers, allowing us to hone in on specific drummer’s styles and generate specific drummer profiles for analysis.

While the original Gillick et al. (2019) article used neural network representations for generative drum models, this study aims at providing interpretable outputs and intuitive analysis through more traditional statistical techniques. This study employs three complementary data mining approaches: dimension reduction (PCA) to identify global stylistic dimensions, kernel density estimation (KDE) to characterize distributional signatures, and graphical models (HMM) to explore temporal structure within performances.

Overall, this study asks i) What are the latent dimensions that best capture variation in drummer playing styles across drum voice usage patterns, timing, and velocity (dynamics)? and ii) Do individual drummers exhibit distinct velocity (dynamics) and timing distributions that can be characterized and distinguished using kernel density estimation?

Literature Review

The dataset used in this study (the Groove MIDI Dataset (GMD)) comes from a paper by Gillick et al. (2019) that was developed with the goal of generating human-like drum sequences with neural networks. Specifically, the group applied the technique of inverse sequence transformations which is, like in the name, an inversion of sequence transformation. Breaking this down, performance data is often transformed into quantifiable data with different “voices” (pitch variations) removed, helping individuals model the general patterns from a pure sound perspective. This is known as sequence transformation. Gillick et al. (2019) constructed a model that takes transformed performance data and inverts it back to its original state. The goal of this approach is to take a dataset of performances and regenerate new, human-like performances. Within the neural network construction, the researchers created “groove embeddings” using variational information bottleneck encoder (VIB) and other latent-space architectures. These “groove embeddings” capture rich feature information not easily engineered. This becomes important in designing an accurate neural network that can successfully take the complexities of a particular drummer’s performance and generate a synthetic, yet human-like performance. While this research was important, it does not answer the questions outlined in the introduction of this study because the “deep embeddings” that neural networks are architected from are not interpretable features; if they were, then anyone could come up with them in a feature engineering phase. For interpretable features of performance data, it’s more advantageous to apply more traditional data mining approaches based on a known feature-set.

An interesting extension to the Gillick et al. (2019) article is that of the Expanded Groove MIDI Dataset (E-GMD), introduced by Callender et al. (2020), which is considerably larger than the original GMD. This new dataset, in addition to including considerably more MIDI recordings, also include audio data in the form of waveforms. This study, while providing more data than necessary for this project, contributes in its key finding: velocity empirically matters for drumming style and perceived performance quality. Thus, this study include velocity distributional-analysis.

Other studies emphasized the statistical insights that can be gained from studying drumming more generally. For example, in “Mapping timing strategies in drum performance”, Sioros, Camera & Danielsen (2019) studied 22 professional drummers performing a common drum pattern with “Pushed”, “Laid-back”, and “On” tempo instructions and compared. From there, they extracted micro-timing features to build a “microtiming profile” per performance. They applied hierarchical clustering to these profiles and found that drummers exhibit unique signatures of time, supporting this study’s hypothesis that drummers exhibit specific, quantifiable playing styles. This idea of specific “microtiming profiles” was further developed when Senn (2023) proposed a model based on predictive-coding/surprisal where the perceived complexity of a drum pattern equals how much it surprises the listener’s internal predictive model (what the listener implicitly anticipates should come next). The model reached an R^2 of 0.852, providing evidence that statistical features such as timing and dynamics can map to perceptual style features like complexity. Moreover, Danielson et al. (2024) also found that not only do timing and dynamics matter for drumming performance perception, but also what sounds are produced. Once again examining the microstructure of sound changes, the researchers showed that timbre, attack, and articulation are also important in perception. Overall, these three studies provide evidence that the perceived timing and feel of drumming is multi-dimensional, not simply based on its adherence to the prescribed tempo. Additionally, the relationship between drumming style and traditional genre labels requires further analysis. This supports this study’s emphasize on a multi-dimensional feature set that includes velocity, timing, and pitch.

Overall, while Gillick et al. (2019) demonstrated that the GMD contains sufficient information to model drumming style through neural embeddings, no prior work exists that provides an interpretable feature-set for drumming style identification, especially on this data. Thus, the literature supports this study’s uniqueness and utility in providing interpretable insights as well as this study’s feasibility, given the progress made more generally in drumming analysis.

Methods

This study analyzes the Groove MIDI Dataset (Gillick et al., 2019), comprising 446,312 drum hits from 1,150 MIDI recordings across 10 professional drummers. One important item to note is that each drummer contributes unequal numbers of recordings (9-494) and hits (2,227-116,019) to the total dataset (see Appendix C, Tables 14-15 & Figures S1-S2 for details). Despite this imbalance, standard errors for all summary statistics remain negligible even for the smallest sample ($SE < 1$ for mean velocity), ensuring reliable feature estimates across all drummers.

To address the first research question through PCA, this study engineers a 32-dimensional feature space (31 features used for PCA) comprised of velocity, timing, drum voice, and hit density features. These features are aggregated across all recordings for each drummer, capturing their overall playing style. Specifically, they include 5 **velocity features** capturing mean, standard deviation, median, range, and coefficient of variation. Additionally, there are 4 **timing features** capturing mean, standard deviation, median, and coefficient of variation of inter-onset intervals. Next, are the 22 **drum voice usage** features capturing the proportional usage of each MIDI pitch (drum type). Finally, there is an added **hit density** feature that captures just how many hits are performed by each drummer across all their individual recordings, but is not used in PCA because of imbalance in the hit-count per drummer and recording-count per drummer.

Now that the 32-dimensional feature space has been defined, PCA was applied to the data with these features (excluding `total_hits`) to identify the latent dimensions of playing style. As is customary with PCA, the features were standardized using z-score normalization, defined as $z_i = \frac{x_i - \mu}{\sigma}$, where x_i = the original value of the feature for observation i , μ = the sample mean of that feature, and σ = the sample standard deviation of that feature. This ensures that the features are all given equal weight.

For each drummer, KDE was applied to velocity and inter-onset interval distributions using Gaussian kernels. The choice to use Gaussian kernels stems from their computational efficiency, theoretical optimality for smooth continuous data, and widespread use in density estimation applications. Notably, KDE was applied on all individual hits grouped by drummer rather than the aggregated features in the 31-dimensional feature space defined for PCA. This reveals the full distributional shape for each drummer, suggesting their overall tendencies and determine if certain drummers exhibit multimodal distributions (suggesting discrete dynamic levels) or unimodal distributions (suggesting continuous variation). The bandwidth was selected using Silverman’s rule of thumb: $\hat{h} = 1.06\hat{\sigma}n^{-1/5}$, where $\hat{\sigma}$ is the sample standard deviation and n is the sample size.

After conducting PCA to identify latent stylistic dimensions and KDE to identify the modality of the velocity and inter-onset interval distributions for each drummer, this study extends to a brief exploration into the temporal structure of velocity for each recording per drummer. This can reveal further nuances surrounding drummer style. For brevity and efficiency and in recognition that some drummers possess more recordings than others (Appendix C, Table 15), this study only considers one representative recording per drummer. For consistency, this study selects rock as

the genre for each recording. To examine the temporal structure of each recording, this study fit three-state HMMs to each recording, modeling the velocity sequence $V_1, V_2, \dots, V_T$ as observations from latent dynamic states $S_t \in \{1, 2, 3\}$ (soft, medium, loud), with:

- **Emission Distributions:** $P(V_t|S_t = k) = \mathcal{N}(\mu_k, \sigma_k^2)$ (Gaussian)
- **Transition Probabilities:** $P(S_{t+1} = j|S_t = i) = A_{ij}$
- **Initial State Distribution:** $P(S_1 = k) = \pi_k$

Furthermore, parameters are estimated using a specific case of the general expectation maximization (EM) algorithm called the Baum-Welch algorithm (implemented in the `depmixS4` package). This estimation method results in characterizations of **state means** μ_k (discrete dynamic levels identified by the model), the **transition matrix** A (how drummers move between states over time), and finally, **state persistence** which is the average diagonal values of A (tendency from drummers to stay in same state).

Statistical Inference

Table 1: Data Distribution per Drummer

Drummer	Recordings	Hits	% Rec.	% Hits
drummer1	494	116,019	43.0	26.0
drummer7	342	52,009	29.7	11.7
drummer3	98	10,089	8.5	2.3
drummer8	80	111,328	7.0	24.9
drummer5	54	101,211	4.7	22.7
drummer9	30	7,946	2.6	1.8
drummer2	18	26,323	1.6	5.9
drummer6	15	12,354	1.3	2.8
drummer10	10	6,806	0.9	1.5
drummer4	9	2,227	0.8	0.5
Total	1,150	446,312	100.0	100.0

One final note before outlining the results concerns the validity of statistical inference. This study makes no definitive inferential claims because the 10 drummers represent a convenience sample from the Groove MIDI Dataset (Gillick et al., 2019) rather than a probability sample from a defined population (Devore & Berk, 2012). Moreover, the severe imbalance in hits and recording representation as shown in Table 1 above and the nested data structure of hits within recordings per drummer violates the independence assumption when trying to generalize to the population of drummers (Appendix C, Tables 14-15 & Figures S1-S2). Thus, in the spirit of data mining, this study should be treated as hypothesis-generating rather than hypothesis-testing with any results motivating inferential and confirmatory studies.

Results

Principal Component Analysis

Principal Component Analysis (PCA) was applied to the 31-dimensional feature space, revealing that the first three principal components explain 63.8% of total variance (PC1: 27.6%, PC2: 24.8%, PC3: 11.5%; Table 2). The scree plot shows a clear elbow at PC3 (Appendix C, Figure S3), supporting retention of these three components. Moreover, PC4 and beyond show idiosyncratic feature loadings and do not individually contribute much to explaining the variance, with PC4 explaining less than 10% and subsequent components diminishing further. Thus, while 63.8% variance-explained is lower than optimal (~70-85%), only the first three PCs will be kept to preserve interpretability. Drummers exhibit clear separation in PC space (Appendix C, Figure S4; PC scores in Table 16).

Table 2: Variance Explained by Principal Components

Component	Standard Deviation	Proportion of Variance	Cumulative Proportion
PC1	2.922	0.275	0.275
PC2	2.771	0.248	0.523
PC3	1.887	0.115	0.638
PC4	1.732	0.097	0.735
PC5	1.623	0.085	0.820
PC6	1.450	0.068	0.888
PC7	1.270	0.052	0.940
PC8	1.020	0.034	0.973
PC9	0.912	0.027	1.000
PC10	0.000	0.000	1.000

The feature loadings (Table 3) reveal 3 distinct dimensions among the three PCs. The first, PC1, can be thought of as drum-voice usage, defined in this study as “Kit Utilization” because the top 5 loadings are pitch features. The highest loading on PC1 (**prop_pitch_46**) represents the open hi-hat sound and has a 0.303 feature loading. The remaining are **prop_pitch_51** meaning the ride cymbal (0.300), **prop_pitch_48** meaning the hi-mid tom (0.295), **prop_pitch_38** meaning the snare drum (0.272), and **prop_pitch_36** meaning the kick drum (0.270). This reveals that the drummer’s choices of which drum to play capture the majority of the variance in the hits, more so than qualities such as how fast (or subdivided) a drummer plays or how loud (or soft) a drummer plays. The second principal component, PC2, is primarily dominated by velocity and timing features, with **mean_velocity** and **mean_ioi** (mean inter-onset interval or the mean of the “timing” variable) taking the top two spots in terms of feature loading (0.322 and 0.306 respectively). Interestingly, **prop_pitch_45** (the low-tom voice) takes the third feature loading spot (0.291), suggesting that voice patterns are still important, even if not in the first principal component. The features **velocity_cv** and **median_velocity** are also present, with loadings -0.290 and 0.288, respectively. This reinforces that PC2 is primarily focused on power (dynamics) and tempo, with higher scores indicating louder, more consistent playing with slower tempo and lower scores indicating softer, more variable dynamics with faster tempo. Finally, the third principal component, PC3, captures additional hi-hat and sticking voices with **prop_pitch_26** meaning open hi-hat, **prop_pitch_42** meaning closed hi-hat, **prop_pitch_44** meaning pedal hi-hat, and **prop_pitch_37** meaning sidesticking on the snare drum, all with loadings 0.392, 0.323, 0.278, and 0.268, respectively. Another feature of note is that **cv_ioi** is also present in the top 5 loadings for

PC3, suggesting some residual timing complexity that matters. Overall, PCA suggests that drum voice preferences matter the most in this data, quickly followed by timing and power, and then specific rhythmic and musical complexity on the hi-hat being the third most contributive.

Table 3: Top 5 Feature Loadings for PC1–PC3

	Principal Component	Feature	Absolute Loading
PC1			
prop_pitch_46	PC1	prop_pitch_46	0.3031738
prop_pitch_51	PC1	prop_pitch_51	0.3001304
prop_pitch_48	PC1	prop_pitch_48	0.2946264
prop_pitch_38	PC1	prop_pitch_38	0.2721573
prop_pitch_36	PC1	prop_pitch_36	0.2697360
PC2			
mean_velocity	PC2	mean_velocity	0.3219551
mean_ioi	PC2	mean_ioi	0.3062416
prop_pitch_45	PC2	prop_pitch_45	0.2913740
velocity_cv	PC2	velocity_cv	0.2896556
median_velocity	PC2	median_velocity	0.2877453
PC3			
prop_pitch_26	PC3	prop_pitch_26	0.3915051
cv_ioi	PC3	cv_ioi	0.3890949
prop_pitch_42	PC3	prop_pitch_42	0.3225740
prop_pitch_44	PC3	prop_pitch_44	0.2780255
prop_pitch_37	PC3	prop_pitch_37	0.2681252

Kernel Density Estimation

As discussed in the Methods, KDE was applied to individual hits, separated by drummer to reveal the distributions of the variables inter-onset interval and velocity, respectively. The inter-onset interval distributions showed similar patterns of multimodality, with peaks occurring at around 0 seconds. Most secondary modes occurred at around 0.125 seconds and tertiary modes just past 0.25 seconds (Figure 1).

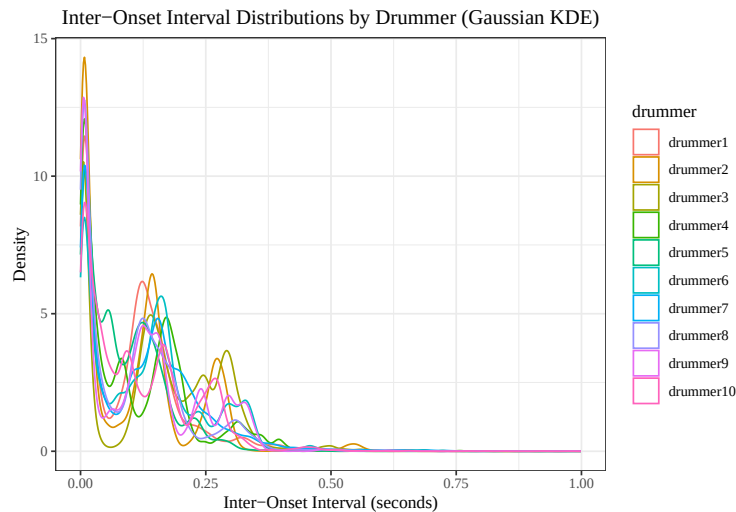


Figure 1: IOI Distributions Overlaid

Interestingly, the velocity distributions exhibit a highly diverse mixture of different multimodal

distributions across drummers (Figure 2). Three groups with similar multimodal trends emerge from this analysis. First, **drummer4**, **drummer7**, and **drummer9** exhibit similar bimodal structures, with gradual peaks in the mid-velocity range (50-100) and spikes at the maximum MIDI velocity (127). Interestingly, **drummer9** and **drummer4** show almost identical modal shapes, with the only difference being that the first peak in **drummer4**'s distribution is higher than the second peak, with a flipped case in **drummer9**. The second group contains **drummer6**, **drummer10**, **drummer8**, and **drummer3** which show similar spikes at velocity 127 and much lower peaks around the mid-range of velocity than with the first group. Finally, the third group contains **drummer2** and **drummer5**, which have trimodal patterns, each with a single, strong middle peak and middle-to-small peaks on either extreme. Additionally, **drummer1**'s distribution, while not trimodal, follows a similar pattern as **drummer2**'s distribution, with a strong central peak and a spike toward one extreme. The consistency in a spike at velocity 127 suggests that all 10 drummers employ accented hits sparingly in their performances. More discussion surrounding the three velocity groups will follow.

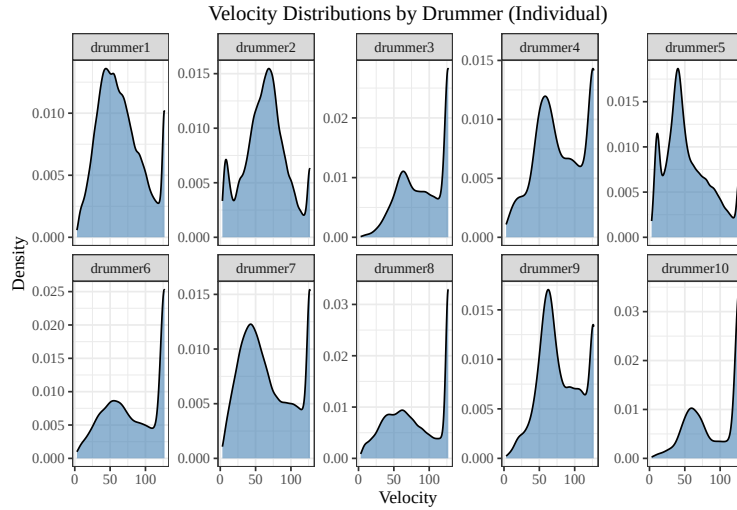


Figure 2: Velocity Distributions by Drummer (Faceted)

Table 4: Integration: PC Scores and Velocity/Timing Characteristics

drummer	PC1	PC2	PC3	vel_mean	vel_sd	vel_range	vel_cv	ioi_mean	ioi_sd	ioi_cv
drummer1	-0.82	-2.06	1.03	64.30	30.61	123	0.48	0.11	0.10	0.91
drummer2	-6.32	-3.40	-1.40	64.10	30.02	124	0.47	0.11	0.11	0.98
drummer3	-2.54	4.36	2.75	91.66	31.46	122	0.34	0.16	0.13	0.80
drummer4	3.94	-0.06	-0.88	80.45	34.91	122	0.43	0.11	0.11	0.97
drummer5	3.52	-4.42	1.85	54.41	31.85	123	0.59	0.09	0.08	0.87
drummer6	0.55	2.30	0.12	84.12	38.04	123	0.45	0.14	0.12	0.83
drummer7	0.38	-0.07	1.20	67.44	36.56	123	0.54	0.13	0.13	0.96
drummer8	0.06	-0.83	-1.05	80.22	37.71	123	0.47	0.11	0.11	0.98
drummer9	0.05	2.11	0.21	79.49	30.10	122	0.38	0.13	0.12	0.92
drummer10	1.18	2.08	-3.83	94.83	34.55	123	0.36	0.12	0.12	0.99

To understand how PCA dimensions may correspond to distributional characteristics in the velocity and inter-onset interval features, the table above (Table 4) shows the correlation coefficients between PC scores and the KDE-derived statistics. These correlations serve as descriptive statistics describing the strength of the association within the 10 drummers and as stated in the Methods,

do not make any inferential claims (hence, no p-values). PC2 shows a very strong positive correlation with mean velocity ($r=0.89$) and strong negative correlation with velocity CV ($r=-0.80$) which confirms the power (dynamics) portion of the power and timing dimension (Table 5). Additionally, in Table 4, PC3 shows strong negative correlation with IOI CV ($r=-0.73$), suggesting that hi-hat voicing is more prominent in areas of timing variability (`cv_ioi` represents timing variability, not pure speed). PC1 shows negligible correlation with mean velocity ($r=0.03$), confirming drum voice selection is independent of dynamic contrast (Table 5).

Table 5: Correlations Between PC Scores and Velocity/Timing Statistics

Relationship	Correlation (r)
PC2 vs Mean Velocity	0.892
PC2 vs Velocity CV	-0.803
PC3 vs IOI CV	-0.734
PC1 vs Mean Velocity	0.030

Hidden Markov Models

While both the velocity distributions and inter-onset interval distributions showed interesting patterns, this extension focuses on applying Hidden Markov Models (HMM) to only the velocities for each drummer. The rationale here is that the velocity distributions showed more diversity in modal shapes and trends than the inter-onset interval distributions, so a temporal analysis may be more fruitful. The HMMs fit to representative rock recordings identified discrete dynamic states with means ranging from 20-50 (soft) to 80-110 (loud). State persistence varies across drummers (range: 0.25-0.70, mean: 0.40 ± 0.16), with some drummers showing high diagonal stability (drummer2: 0.70, drummer4: 0.65) while others exhibit more frequent state transitions (Table 6). This variation suggests individual differences in dynamic stability, though all drummers demonstrate sustained periods within states rather than rapid fluctuations.

Table 6: HMM Characteristics: Rock Performances

	Drummer	Hits	Soft	Medium	Loud	Persistence	Switching
drummer1	drummer1	226	48.9	93.8	127.0	0.30	0.35
drummer2	drummer2	9282	8.0	50.9	76.0	0.70	0.15
drummer3	drummer3	444	71.1	108.0	127.0	0.25	0.38
drummer4	drummer4	420	49.8	72.5	127.0	0.65	0.18
drummer5	drummer5	268	40.1	94.6	101.4	0.26	0.37
drummer6	drummer6	243	72.1	74.4	127.0	0.35	0.33
drummer7	drummer7	221	41.3	81.8	102.1	0.33	0.34
drummer8	drummer8	223	76.7	97.3	127.0	0.31	0.35
drummer9	drummer9	337	64.7	70.5	119.6	0.44	0.28

Now that the HMMs have been fit and examined, below is an examination of whether HMM-derived characteristics for velocity relate to the dimensions and distributions identified earlier.

Table 7: HMM Integration with PCA and KDE Results

drummer	PC2	vel_mean	vel_cv	loud_state	persistence
drummer1	-2.06	64.30	0.48	127.0	0.30
drummer2	-3.40	64.10	0.47	76.0	0.70
drummer3	4.36	91.66	0.34	127.0	0.25

drummer4	-0.06	80.45	0.43	127.0	0.65
drummer5	-4.42	54.41	0.59	101.4	0.26
drummer6	2.30	84.12	0.45	127.0	0.35
drummer7	-0.07	67.44	0.54	102.1	0.33
drummer8	-0.83	80.22	0.47	127.0	0.31
drummer9	2.11	79.49	0.38	119.6	0.44
drummer10	2.08	94.83	0.36	NA	NA

Instead of outputting the transition matrix for all 10 drummers (Appendix C, Figure S7), the figure below displays the transition matrix for **drummer1** as a representative example of state-switching behavior. There are two key areas to note: diagonal and non-diagonal values. Diagonal values indicate sustained state periods whereas non-diagonal values indicate transition periods. For **drummer1**, we actually do not see high diagonal values across the matrix. We only see a high diagonal value at medium-to-medium (0.78) and we see high values at soft-to-medium (0.78) and loud-to-medium (0.86). This conflicts with our global observation that the 10 drummers typically remain in specific dynamic levels for sustained periods of time (for rock recordings). Nevertheless, these values still indicate that **drummer1** is less erratic in dynamic style, instead, playing this particular rock recording with crescendos and diminuendos that modulate around the medium dynamic level. Thus, while the transition matrix for **drummer1** does not confirm the global observation exactly, it still provides evidence for smooth, consistent playing.

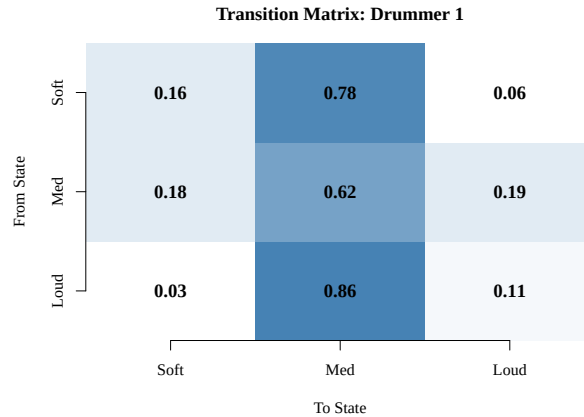


Figure 3: Transition Matrix: Drummer1

Discussion

Beginning with the first posed research question, PCA reduced down to 3 optimal principal components capturing 63.8% of the variation in drumming style. The top 5 loadings of PC1 were all pitch features, providing evidence that drum voice selection explains 27.6% of the variance (slightly more than PC2's 24.8%), though the difference is modest (2.8 percentage points). Specifically, PC1 identified that the 10 drummers vary in how they choose to use the open hi-hat sound for accents and the ride cymbal for sustained subdivisions on the high end of the pitch register, along with snare and kick drum usage. This is not surprising, as drummers typically spend the most time playing the hi-hat, snare drum, kick drum, and ride cymbals for the majority of their grooves. Therefore, it makes sense why those voices account for the greatest amount of variation in playing

across these 10 drummers. Interestingly, though, the hi-mid tom voice was also one of the top 5 feature loadings for PC1. This result is surprising as tom usage (and hi-mid tom specifically) is not typically thought of as a main differentiator of playing style because drummers typically use those voices for brief drum fills or specific bridge sections, not main grooves. It is perhaps the case that Latin grooves, which typically involves more tom accents, contributed to this, prompting further research into Latin-specific recordings.

Unlike the brief surprise in PC1, PC2 was largely unsurprising, with power and timing being important differentiators. The high-PC2 drummers play more consistently with slower tempo (heavier feel), play softer with more dynamic variation and faster tempo (lighter feel). Notably, the negative correlation in PC2 with `velocity_cv` indicates the more powerful (louder) players are more consistent in their dynamics. Finally, the appearance of the low-tom voice feature in PC2 suggests that voicing remains important in differentiating playing styles in these 10 drummers, even at PCs past PC1.

Finally, PC3 offers a new perspective that speaks to what drummers often call “the pocket”. Once again, a voicing-dominant component, PC3 showed primarily hi-hat voicing and timing consistency, with a key finding being a negative correlation with `ioi_cv`, indicating that more complex hi-hat articulation is associated with more consistent timing. This is characteristic of drummers with a keen sense of “the pocket”, as it colloquially describes drummers who employ complex rhythms with few voices and consistent timing throughout. Moreover, when drummers refer to “the pocket”, they’re specifically referring to modulating around the hi-hat, snare, and bass drum which were the voices captured by PC3. Overall, PCA showed that differences between drummers arise from three independent dimensions defined by kit utilization, power/tempo, and timing articulation, with kit utilization and power/tempo being nearly equally important.

Moving onto the second research question, the KDE results revealed that drummers do exhibit distinct velocity and timing distributions that serve as reliable “signatures”. Timing distributions remained largely consistent, with all being multimodal with similar patterns of modality (trimodal). This reflects individual rhythmic spacing preferences among the 10 drummers. Moreover, the more uniform structure of IOI distributions compared to velocity distributions motivated focusing extended validation efforts on velocity characteristics. The velocity distributions revealed diversity in drummer choices among the 10 drummers, with three primary distributional types. The first type, the bimodal structures, show a group of drummers that concentrate their playing around the medium dynamic levels. The second type showed drummers who play with accents on the loudest dynamic end, and less frequently on the medium-softer dynamic range. The third type showed a group of drummers that play primarily on the medium dynamic range, but with departures on the extremes (maximally loud accents and hits on the softest dynamic end). As shown in the Results, **drummer5** stands apart with a primarily unimodal distribution, showing a dynamically consistent playing style. Overall, the KDE distributional signatures reveal that individual drummers do exhibit distinct, quantifiable patterns that can be characterized through KDE. Moreover, the strong integration between PCA and KDE ($r_{PC2-to-velocity} = 0.89$) validates both approaches: PCA captures global dimensions, while KDE reveals detailed distributional structure.

The extension to HMM proved insightful, revealing how the multimodal patterns identified by KDE manifest temporally between dynamics states (the velocity features). HMMs successfully identified 3 discrete dynamic states (soft, medium, loud) in rock recordings, aligning with the modes visible in KDE plots. Furthermore, the transition matrices revealed varied temporal dynamics across drummers. In the investigation of **drummer1**, the transition matrix showed that this drummer modulates around the medium dynamic level, choosing to move gradually between adjacent dynamic

levels rather than jumping from soft-to-loud and loud-to-soft. While the KDE results indicated that the diagonal entries (same-state transitions) would dominate, this result still communicates that, for `drummer1`, the playing style is consistent and smooth, rather than erratic. This extends the KDE results to communicate not simply the distribution of dynamic levels, but provides insight into the decisions and movement of the 10 drummers. Overall, this extension demonstrates that KDE (distributional patterns) and HMM (temporal structure) provide complementary perspectives on the dynamic styles of these 10 drummers.

Limitations

This study has several limitations. As discussed in the Methods section, this study does not include inferential claims as the 10-drummer dataset is convenience sample rather than a probability sample. Moreover, the severe imbalance in the data would violate independence assumptions required for statistical inference. Thus, this study is limited in that it only provides hypothesis-generating findings rather than hypothesis-testing, confirmatory analysis. Another limitation is that the identities of these drummers are not disclosed, making it difficult to determine if these are studio drummers with a wide repertoire of playing styles or more famous drummers with specific “native” playing styles. As a result, this analysis is limited in its ability to confirm whether or not the drumming style differences are from true individual differences or based more on the background of the drummer. A limitation in the HMM analysis was that it was only ran on one rock recording per drummer which limits the individual analysis. As a result of the imbalance in the data, applying HMM to every recording would not necessarily accurately reveal the drumming styles since different drummers have different recording counts for each genre. Moreover, applying HMM to every recording would be computationally expensive, as that would require fitting 1,150 HMMs. Finally, PCA and KDE aggregate across all recordings, potentially obscuring within-drummer variability across different musical contexts.

Future Work

As this study is hypothesis-generating, there are several directions for future research. Expanding the HMM analysis to multiple recordings per drummer across genres would provide for a more robust analysis of temporal dynamics. This analysis would determine whether transition patterns remain stable across musical contexts. Additionally, inclusion of the Expanded Groove MIDI Dataset (E-GMD) could expand the temporal analysis to waveform analysis, potentially revealing at a granular level, what the playing style variations are and where they come from in the course of a groove. Another useful analysis, if given a larger, more representative dataset along with additional demographic metadata on the drummers themselves, is genre analysis. Comparing drummer style to genre labels and accounting for drummer background could yield more developed inferential conclusions surrounding whether colloquial genre labels match the data.

Conclusion

In conclusion, this study sets the foundation for interpretable characterizations of drumming styles through modern data mining methods. The complimentary application of principal component analysis and kernel density estimation addresses the two research questions: i) the latent dimensions (63.8% variance captured) were determined to be kit utilization (PC1), power/tempo (PC2), and timing articulation (PC3) and ii) this study found that these 10 drummers exhibit distinct velocity patterns with one group exhibiting more bimodal mid-range structures, another group exhibiting accent-dominated distributions, and a final group with trimodal patterns with central dominance in the mid-range. The integration of PCA and KDE showed strong correlations with the power/timing dimension and KDE velocity signatures and HMM loud state parameters, showing consistent global characteristics in the power/timing dimension. Furthermore, the extension to HMM proved to be a useful alternate perspective on the velocity distributional signatures, with the transition matrices refining the drumming style conclusions made after KDE. Moreover, the convergence of these state-space transition probabilities suggests that these characteristics identified in KDE and PCA are more stable than previously evident.

As a descriptive, hypothesis-generating study of 10 drummers, these findings establish interpretable, quantitative methods for characterizing musical performance style that complement existing neural approaches. By revealing that individual stylistic characteristics manifest consistently across multiple analytical perspectives (from global feature summaries to distributional shapes to temporal transition patterns) this work provides a framework for future confirmatory studies with larger, probability-based samples that could support statistical inference about broader populations of drummers. This work also supports an extension into genre analysis, exploring HMMs in different genres.

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Appendix A: Data Dictionary

Variable	Variable Name	Type	Allowed Values	Description
Drummer ID	drummer	String	drummer1-drummer10	Anonymous identifier for the drummer who performed the recording
Session ID	session	String	Any string	Unique identifier for the recording session
Performance ID	id	String	Any string	Unique identifier for the individual performance/recording
Genre	style	String	Any string	Musical genre/style label (e.g., jazz, rock, funk, latin)
Tempo	bpm	Numeric	Any positive number	Beats per minute of the recording
Beat Type	beat_type	String	Any string	Type of beat pattern (e.g., beat, fill)
Time Signature	time_signature	String	Format: N/D	Time signature of the recording (e.g., 4/4, 3/4, 6/8)
Time	time	Numeric	Any non-negative number	Timestamp of the drum hit in seconds from start of recording
Velocity	velocity	Integer	0-127	MIDI velocity value indicating how hard the drum was hit (dynamics)
Pitch	pitch	Integer	22-59	MIDI pitch number corresponding to specific drum voice (e.g., 36=kick, 38=snare, 42=closed hi-hat)

Variable	Variable Name	Type	Allowed Values	Description
Inter-Onset Interval	inter_onset_interval	Numeric	Any non-negative number	Time in seconds between consecutive drum hits $IOI_i = t_{i+1} - t_i$

MIDI Drum Map (General MIDI Standard)

The dataset uses the General MIDI (GM) drum map standard, where each MIDI pitch number corresponds to a specific drum or cymbal in a standard drum kit:

MIDI Pitch	Drum/Cymbal	Category
22	Closed Hi-Hat (foot)	Hi-Hat
26	Open Hi-Hat (alternate)	Hi-Hat
36	Kick Drum (Bass Drum)	Drums
37	Side Stick / Rimshot	Snare
38	Acoustic Snare	Drums
40	Electric Snare	Drums
42	Closed Hi-Hat	Hi-Hat
43	High Floor Tom	Toms
44	Pedal Hi-Hat	Hi-Hat
45	Low Tom	Toms
46	Open Hi-Hat	Hi-Hat
47	Low-Mid Tom	Toms
48	Hi-Mid Tom	Toms
49	Crash Cymbal 1	Cymbals
50	High Tom	Toms
51	Ride Cymbal 1	Cymbals
52	Chinese Cymbal	Cymbals
53	Ride Bell	Cymbals
55	Splash Cymbal	Cymbals
57	Crash Cymbal 2	Cymbals
58	Vibraslap	Effects
59	Ride Cymbal 2	Cymbals

Appendix B: Engineered Features Dictionary

To apply dimensionality reduction, this study aggregated the 446,312 individual drum hits into 32 summary features per drummer (31 used for PCA). These features characterize each drummer’s overall playing style across all recordings and are standardized before PCA.

Velocity Features (n = 5)

These features capture the dynamics (loudness) and consistency of hitting:

Feature	Formula	Interpretation
mean_velocity	$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$	Average hit intensity across all hits
sd_velocity	$s_v = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (v_i - \bar{v})^2}$	Variability in hit intensity
median_velocity	$\text{median}(v_1, \dots, v_n)$	Typical hit intensity (robust to outliers)
velocity_range	$\max(v_i) - \min(v_i)$	Span from softest to loudest hit
velocity_cv	$CV_v = \frac{s_v}{\bar{v}}$	Relative variability (coefficient of variation)

Timing Features (n = 4)

These features capture rhythm, tempo, and timing consistency:

Feature	Formula	Interpretation
mean_ioi	$\bar{t} = \frac{1}{n-1} \sum_{i=1}^{n-1} (t_{i+1} - t_i)$	Average time between consecutive hits (seconds)
sd_ioi	$s_t = \sqrt{\frac{1}{n-2} \sum_{i=1}^{n-1} (IOI_i - \bar{t})^2}$	Variability in inter-onset intervals
median_ioi	$\text{median}(IOI_1, \dots, IOI_{n-1})$	Typical spacing between hits
cv_ioi	$CV_t = \frac{s_t}{\bar{t}}$	Relative timing variability

Drum Voice Usage Features (n = 22)

These features capture timbre preferences and drum kit utilization. Each feature represents the proportion of hits on a specific drum or cymbal:

Feature	MIDI Pitch	Drum/Cymbal	Formula
prop_pitch_22	22	Closed Hi-Hat (foot)	$\frac{\sum I(p_i=22)}{n}$
prop_pitch_26	26	Open Hi-Hat (alt)	$\frac{\sum I(p_i=26)}{n}$
prop_pitch_36	36	Kick Drum	$\frac{\sum I(p_i=36)}{n}$
prop_pitch_37	37	Side Stick	$\frac{\sum I(p_i=37)}{n}$
prop_pitch_38	38	Snare Drum	$\frac{\sum I(p_i=38)}{n}$
prop_pitch_40	40	Electric Snare	$\frac{\sum I(p_i=40)}{n}$
prop_pitch_42	42	Closed Hi-Hat	$\frac{\sum I(p_i=42)}{n}$
prop_pitch_43	43	High Floor Tom	$\frac{\sum I(p_i=43)}{n}$

Feature	MIDI Pitch	Drum/Cymbal	Formula
prop_pitch_44	44	Pedal Hi-Hat	$\frac{\sum I(p_i=44)}{n}$
prop_pitch_45	45	Low Tom	$\frac{\sum I(p_i=45)}{n}$
prop_pitch_46	46	Open Hi-Hat	$\frac{\sum I(p_i=46)}{n}$
prop_pitch_47	47	Low-Mid Tom	$\frac{\sum I(p_i=47)}{n}$
prop_pitch_48	48	Hi-Mid Tom	$\frac{\sum I(p_i=48)}{n}$
prop_pitch_49	49	Crash Cymbal 1	$\frac{\sum I(p_i=49)}{n}$
prop_pitch_50	50	High Tom	$\frac{\sum I(p_i=50)}{n}$
prop_pitch_51	51	Ride Cymbal	$\frac{\sum I(p_i=51)}{n}$
prop_pitch_52	52	Chinese Cymbal	$\frac{\sum I(p_i=52)}{n}$
prop_pitch_53	53	Ride Bell	$\frac{\sum I(p_i=53)}{n}$
prop_pitch_55	55	Splash Cymbal	$\frac{\sum I(p_i=55)}{n}$
prop_pitch_57	57	Crash Cymbal 2	$\frac{\sum I(p_i=57)}{n}$
prop_pitch_58	58	Vibraslap	$\frac{\sum I(p_i=58)}{n}$
prop_pitch_59	59	Ride Cymbal 2	$\frac{\sum I(p_i=59)}{n}$

Hit Density Feature (n = 1)

Feature	Formula	Interpretation
total_hits	n	Total number of hits across all recordings

Appendix C: Supplemental Exploratory Data Analysis

This appendix contains additional visualizations that support but are not essential to the main narrative.

Table 14: Number of Hits per Drummer

drummer	n
drummer1	116019
drummer8	111328
drummer5	101211
drummer7	52009
drummer2	26323
drummer6	12354
drummer3	10089
drummer9	7946
drummer10	6806
drummer4	2227

Table 15: Number of Recordings per Drummer

Drummer	Recordings
drummer1	494
drummer7	342
drummer3	98
drummer8	80
drummer5	54
drummer9	30
drummer2	18
drummer6	15
drummer10	10
drummer4	9

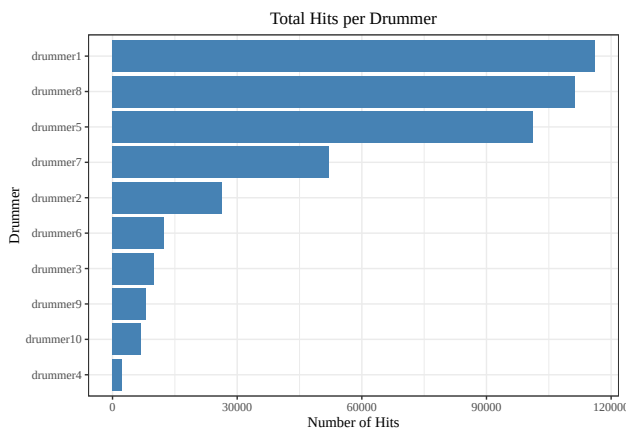


Figure 4: (S1) Total Hits per Drummer

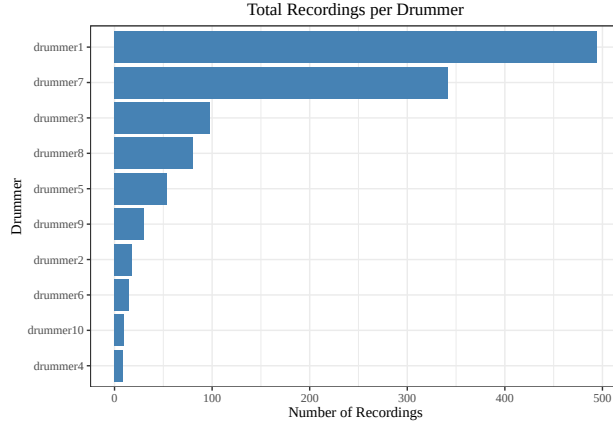


Figure 5: (S2) Total Recordings per Drummer

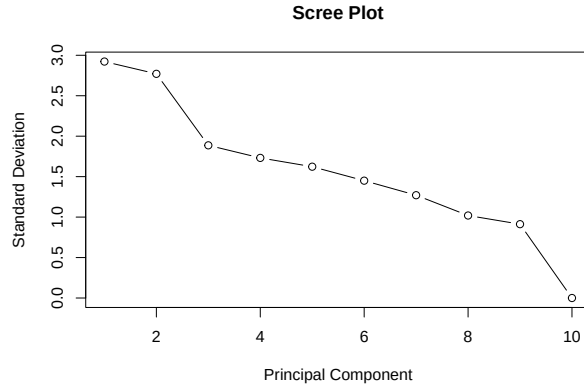


Figure 6: (S3) Scree Plot

Table 16: Principal Component Scores by Drummer

	drummer	PC1	PC2	PC3
drummer1	drummer1	-0.82	-2.06	1.03
drummer2	drummer2	-6.32	-3.40	-1.40
drummer3	drummer3	-2.54	4.36	2.75
drummer4	drummer4	3.94	-0.06	-0.88
drummer5	drummer5	3.52	-4.42	1.85
drummer6	drummer6	0.55	2.30	0.12
drummer7	drummer7	0.38	-0.07	1.20
drummer8	drummer8	0.06	-0.83	-1.05
drummer9	drummer9	0.05	2.11	0.21
drummer10	drummer10	1.18	2.08	-3.83

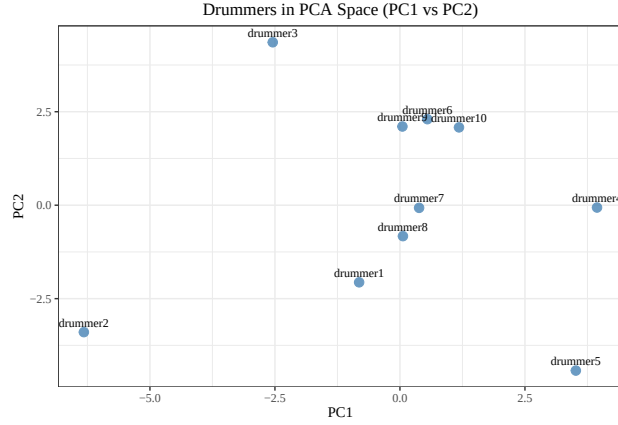


Figure 7: (S4) Drummers in Principal Component Space

Table 17: Feature Loadings on First 3 PCs

	PC1	PC2	PC3
mean_velocity	0.010	-0.322	-0.194
sd_velocity	0.141	-0.082	-0.129
median_velocity	-0.020	-0.288	-0.261
velocity_range	-0.160	0.224	-0.172
velocity_cv	0.082	0.290	0.153
mean_ioi	-0.132	-0.306	0.139
sd_ioi	-0.148	-0.279	-0.056
median_ioi	-0.202	-0.232	0.194
cv_ioi	0.002	0.131	-0.389
prop_pitch_22	0.148	-0.091	0.002
prop_pitch_26	0.129	-0.132	-0.392
prop_pitch_36	0.270	-0.099	-0.063
prop_pitch_37	0.142	0.203	0.268
prop_pitch_38	0.272	0.070	-0.011
prop_pitch_40	0.132	0.063	0.022
prop_pitch_42	-0.072	-0.105	0.323
prop_pitch_43	0.115	-0.134	-0.265
prop_pitch_44	-0.073	0.259	-0.278
prop_pitch_45	0.004	-0.291	0.176
prop_pitch_46	-0.303	0.064	0.024
prop_pitch_47	0.256	0.147	0.080
prop_pitch_48	-0.295	0.054	-0.067
prop_pitch_49	-0.227	0.117	-0.147
prop_pitch_50	0.259	-0.010	0.081
prop_pitch_51	-0.300	-0.017	0.072
prop_pitch_52	0.110	-0.015	0.100
prop_pitch_53	0.131	-0.059	0.088
prop_pitch_55	0.110	-0.276	-0.097
prop_pitch_57	-0.262	0.156	-0.128
prop_pitch_58	0.239	0.086	-0.097
prop_pitch_59	-0.022	-0.099	0.051

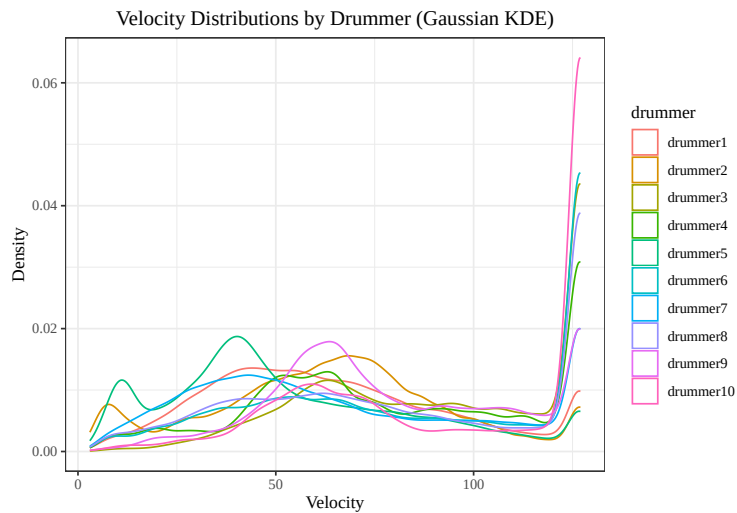


Figure 8: (S5) Velocity Distributions Overlaid

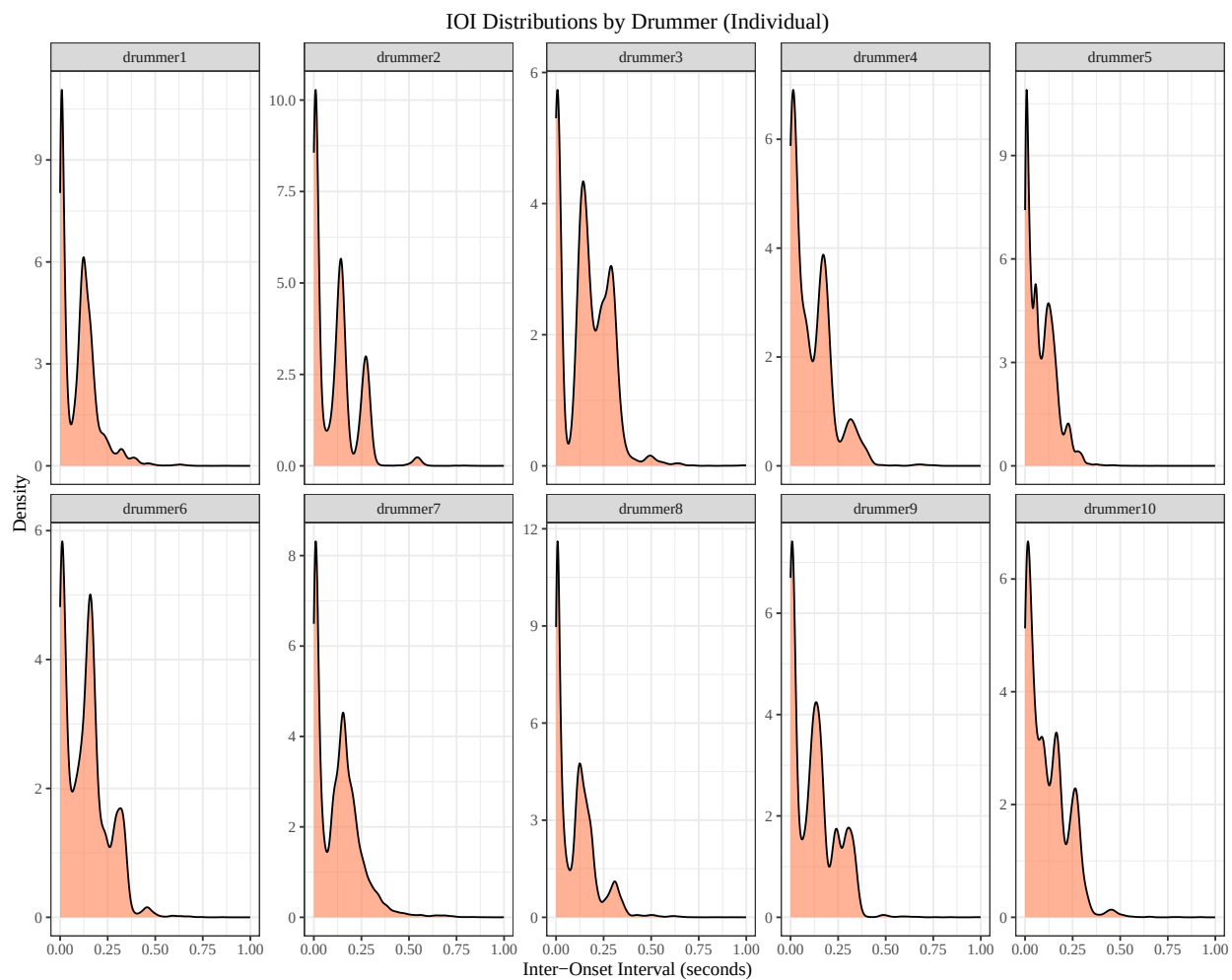


Figure 9: (S6) IOI Distributions by Drummer (Faceted)

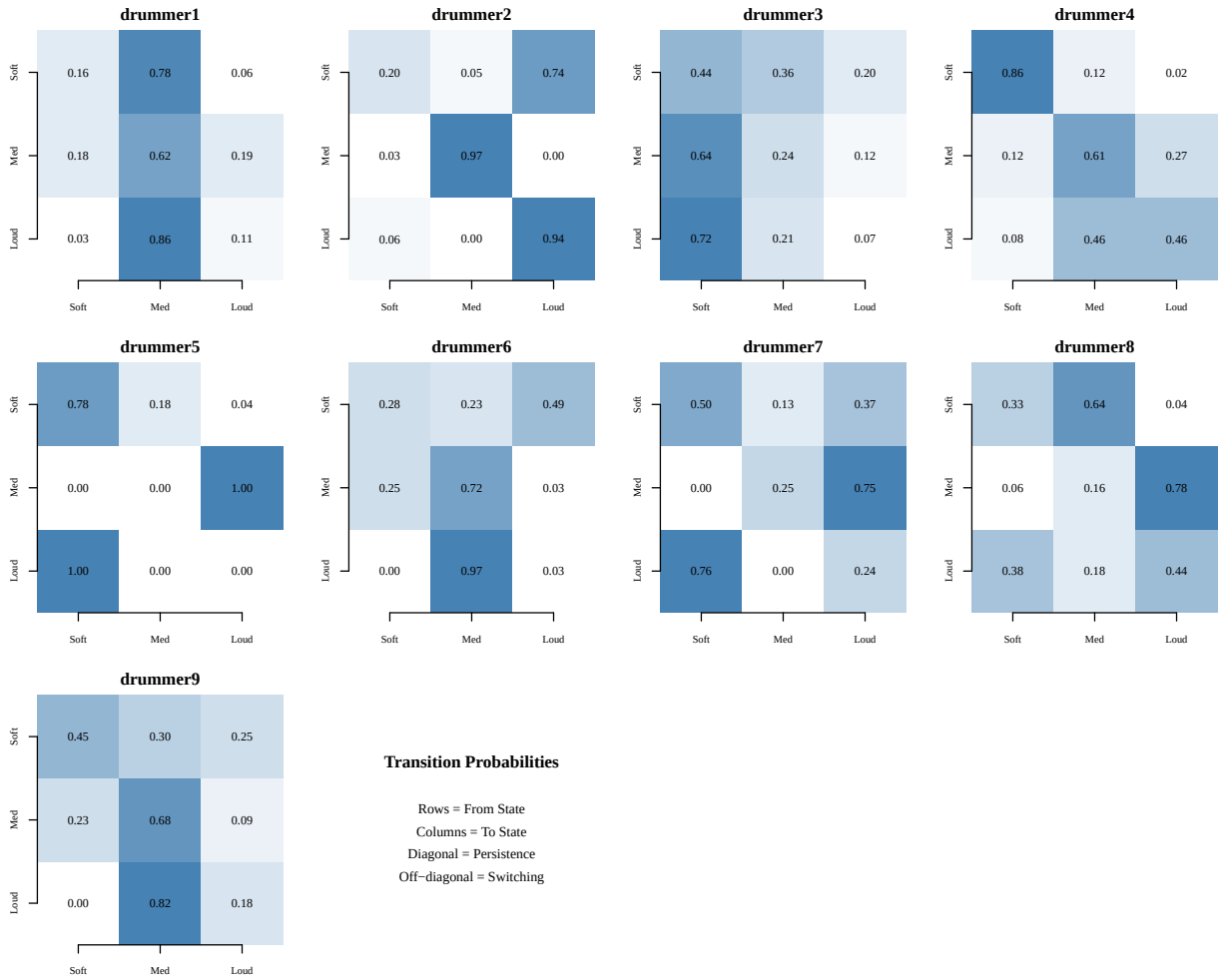


Figure 10: (S7) Complete Transition Matrices for All Drummers

Appendix D: Code

The main data load from the Groove MIDI dataset took place in the Spyder IDE using Python. The decision to move this phase of the study to Python stems from the enhanced capabilities for loading MIDI data, a data form that is often tricky to load. The Groove MIDI dataset is downloadable as the zip file `groove-v1.0.0-midionly` and contains folders for each groove, then folders for each drummer and finally, the MIDI csvs. As such, the Python function below provides an efficient way of reading in the individual files, concatenating into a final csv for reading into RStudio. Note that the filepaths are computer-specific and raw filepaths, so the function may need modification depending on the local directory capabilities of the user.

```
import pandas as pd
import mido

# common path from every file
base_path="C:/Users/naili/OneDrive/Desktop/STAT 5330/Final Project/groove-v1.0.0-midionly/groove"
metadata = pd.read_csv(base_path + "info.csv")
print(f"Found {len(metadata)} MIDI files to process\n")

# Extracting all hits from all files
all_hits = []

for idx, row in metadata.iterrows():
    if idx % 50 == 0:
        print(f"Processing file {idx}/{len(metadata)}...")

# attaching every specific file name to the base path
try:
    midi_path = base_path + row['midi_filename']
    midi_file = mido.MidiFile(midi_path)

    current_time = 0

    for msg in midi_file:
        current_time += msg.time
        if msg.type == 'note_on' and msg.velocity > 0 and msg.channel == 9:
            all_hits.append({
                'drummer': row['drummer'],
                'session': row['session'],
                'id': row['id'],
                'style': row['style'],
                'bpm': row['bpm'],
                'beat_type': row['beat_type'],
                'time_signature': row['time_signature'],
                'time': current_time,
                'velocity': msg.velocity,
                'pitch': msg.note
            })
```

```

except Exception as e:
    print(f"Error processing {row['midi_filename']}: {e}")

# Creating DataFrame
print("\nCreating DataFrame...")
df = pd.DataFrame(all_hits)

# Sorting and calculate inter-onset intervals
print("Calculating inter-onset intervals...")
df = df.sort_values(['drummer', 'session', 'id', 'time']).reset_index(drop=True)
df['inter_onset_interval'] = df.groupby(['drummer', 'session', 'id'])['time'].diff()

# Taking the final df and converting to a csv for easy R-loading
output_path = "C:/Users/naili/OneDrive/Desktop/STAT 5330/Final Project/all_hits.csv"
df.to_csv(output_path, index=False)

# Here are some system messages to ensure that the data has been fully loaded
print(f"Processed {len(df):,} drum hits")
print(f"Saved to: {output_path}")

print("Hits per drummer:")
print(df['drummer'].value_counts().sort_index())

print("\nSample of data:")
print(df.head(10))

```

The remaining code below is in R and was conducted within this R Markdown document.

```

## Data Preprocessing

### Packages

library(tidyverse)
library(knitr)
library(kableExtra)
library(depmixS4)

### Loading in Groove MIDI

all_hits <- read_csv("all_hits.csv")
all_hits <- all_hits %>%
  mutate(drummer = factor(drummer, levels = paste0("drummer", 1:10)))

cat("Total hits loaded:", nrow(all_hits), "\n\n")

kable(all_hits %>%
  count(drummer) %>%

```

```

    arrange(desc(n)))

cat("\n\nColumn names:\n")
print(colnames(all_hits))

cat("\n\nSample data:\n")
kable(head(all_hits, 10))

cat("\n\nVelocity summary:\n")
print(summary(all_hits$velocity))

cat("\n\nPitch (drum types) present:\n")
print(sort(unique(all_hits$pitch)))

# Number of hits per drummer
kable(all_hits %>%
      count(drummer) %>%
      arrange(desc(n)),
      caption = "Number of Hits per Drummer")

#Number of recordings per drummer
recordings_per_drummer <- all_hits %>%
  group_by(drummer) %>%
  summarise(num_recordings = n_distinct(id)) %>%
  arrange(desc(num_recordings))

#combined Table for display in "Methods" section
kable(all_hits %>%
      group_by(drummer) %>%
      summarise(Recordings = n_distinct(id), Hits = n()) %>%
      mutate(`% Rec.` = round(100 * Recordings / sum(Recordings), 1),
             `% Hits` = round(100 * Hits / sum(Hits), 1)) %>%
      arrange(desc(Recordings)) %>%
      bind_rows(summarise(.,
                          drummer = "**Total**",
                          Recordings = sum(Recordings),
                          Hits = sum(Hits),
                          `% Rec.` = 100.0,
                          `% Hits` = 100.0)),
      caption = "Data Distribution per Drummer",
      col.names = c("Drummer", "Rec.", "Hits", "% Rec.", "% Hits"),
      format.args = list(big.mark = ","),
      align = c("l", "r", "r", "r", "r")
    )

kable(recordings_per_drummer,
      caption = "Number of Recordings per Drummer",

```

```

    col.names = c("Drummer", "Recordings"))

# Hits per drummer bar chart
p1 <- ggplot(all_hits %>% count(drummer), aes(x = reorder(drummer, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  theme_bw() +
  labs(title = "Total Hits per Drummer",
       x = "Drummer",
       y = "Number of Hits") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )
p1

# Recordings per drummer bar chart
p2 <- ggplot(recordings_per_drummer <- all_hits %>%
  group_by(drummer) %>%
  summarise(num_recordings = n_distinct(id)) %>%
  arrange(desc(num_recordings)), aes(x = reorder(drummer, num_recordings), y = num_recordings)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  theme_bw() +
  labs(title = "Total Recordings per Drummer",
       x = "Drummer",
       y = "Number of Recordings") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )
p2

## Feature Engineering

# Per drummer features
drummer_features <- all_hits %>%
  group_by(drummer) %>%
  summarise(
    # Velocity features
    mean_velocity = mean(velocity, na.rm = TRUE),
    sd_velocity = sd(velocity, na.rm = TRUE),
    median_velocity = median(velocity, na.rm = TRUE),
    velocity_range = max(velocity) - min(velocity),
    velocity_cv = sd(velocity) / mean(velocity),
    # Timing features
    mean_ioi = mean(inter_onset_interval, na.rm = TRUE),

```

```

sd_ioi = sd(inter_onset_interval, na.rm = TRUE),
median_ioi = median(inter_onset_interval, na.rm = TRUE),
cv_ioi = sd(inter_onset_interval, na.rm = TRUE) / mean(inter_onset_interval, na.rm = TRUE)
# Density
total_hits = n(),
# Drum voice usage (22 MIDI pitches)
prop_pitch_22 = sum(pitch == 22) / n(),
prop_pitch_26 = sum(pitch == 26) / n(),
prop_pitch_36 = sum(pitch == 36) / n(),
prop_pitch_37 = sum(pitch == 37) / n(),
prop_pitch_38 = sum(pitch == 38) / n(),
prop_pitch_40 = sum(pitch == 40) / n(),
prop_pitch_42 = sum(pitch == 42) / n(),
prop_pitch_43 = sum(pitch == 43) / n(),
prop_pitch_44 = sum(pitch == 44) / n(),
prop_pitch_45 = sum(pitch == 45) / n(),
prop_pitch_46 = sum(pitch == 46) / n(),
prop_pitch_47 = sum(pitch == 47) / n(),
prop_pitch_48 = sum(pitch == 48) / n(),
prop_pitch_49 = sum(pitch == 49) / n(),
prop_pitch_50 = sum(pitch == 50) / n(),
prop_pitch_51 = sum(pitch == 51) / n(),
prop_pitch_52 = sum(pitch == 52) / n(),
prop_pitch_53 = sum(pitch == 53) / n(),
prop_pitch_55 = sum(pitch == 55) / n(),
prop_pitch_57 = sum(pitch == 57) / n(),
prop_pitch_58 = sum(pitch == 58) / n(),
prop_pitch_59 = sum(pitch == 59) / n()
)

kable(drummer_features)

# Velocity dist. by drummer
p3 <- ggplot(all_hits, aes(x = velocity, fill = drummer)) +
  geom_density(alpha = 0.5) +
  theme_bw() +
  scale_fill_viridis_d() +
  labs(title = "Velocity Distributions by Drummer",
       x = "Velocity (0-127)",
       y = "Density") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )
p3

# Inter-onset interval dist. by drummer

```

```

p4 <- all_hits %>%
  filter(!is.na(inter_onset_interval),
         inter_onset_interval > 0,
         inter_onset_interval < 1) %>% # Remove outliers
  ggplot(aes(x = inter_onset_interval, fill = drummer)) +
  geom_density(alpha = 0.5) +
  theme_bw() +
  scale_fill_viridis_d() +
  labs(title = "Inter-Onset Interval Distributions by Drummer",
       x = "Time Between Hits (seconds)",
       y = "Density") +
  theme(legend.position = "bottom") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )
p4

# Pitch dist. by drummer
p5 <- ggplot(all_hits, aes(x = pitch, fill = drummer)) +
  geom_density(alpha = 0.5) +
  theme_bw() +
  scale_fill_viridis_d() +
  labs(title = "Pitch (Drum Type) Distributions by Drummer",
       x = "MIDI Pitch Number",
       y = "Density") +
  theme(legend.position = "bottom") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )
p5

# Time dist. by drummer
p6 <- all_hits %>%
  group_by(drummer) %>%
  mutate(time_normalized = time / max(time)) %>% # Normalize to 0-1
  ggplot(aes(x = time_normalized, fill = drummer)) +
  geom_density(alpha = 0.5) +
  theme_bw() +
  scale_fill_viridis_d() +
  labs(title = "Hit Density Over Time by Drummer",
       x = "Normalized Time (0-1)",
       y = "Density") +
  theme(legend.position = "bottom") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
  )

```

```

    text = element_text(family = "serif"),
  )
p6

## PCA on Drummer Features

#Feature matrix for PCA
X <- drummer_features %>%
  dplyr::select(-drummer, -total_hits) %>%
  as.matrix()

rownames(X) <- drummer_features$drummer

# Running PCA
pca_result <- prcomp(X, scale. = TRUE)

pca_scores <- data.frame(
  drummer = drummer_features$drummer,
  PC1 = pca_result$x[, 1],
  PC2 = -pca_result$x[, 2],
  PC3 = pca_result$x[, 3]
)

# Variance explained
summary(pca_result)

#Variance info from PCA summary
pca_summary <- summary(pca_result)$importance
variance_df <- as.data.frame(t(pca_summary))

# Component name column
variance_df <- data.frame(
  Component = rownames(variance_df),
  variance_df
)
rownames(variance_df) <- NULL

# Table
kable(variance_df,
      digits = 3,
      caption = "Variance Explained by Principal Components",
      col.names = c("Component", "Standard Deviation", "Proportion of Variance", "Cumulative P
    ) %>%
  kable_styling(latex_options=c("hold_position"), font_size=8)

# Top contributors to each PC
top_pc <- function(rotation, pc_index, k = 5) {

```

```

    sort(abs(rotation[, pc_index]), decreasing = TRUE)[1:k]
}

top_pc1 <- top_pc(pca_result$rotation, 1)
top_pc2 <- top_pc(pca_result$rotation, 2)
top_pc3 <- top_pc(pca_result$rotation, 3)

top_loadings_df <- data.frame(
  PC = c(rep("PC1", 5), rep("PC2", 5), rep("PC3", 5)),
  Feature = c(names(top_pc1), names(top_pc2), names(top_pc3)),
  Loading = c(top_pc1, top_pc2, top_pc3)
)

# Table
kable(top_loadings_df, caption = "Top 5 Feature Loadings for PC1-PC3",
  col.names = c("Principal Component", "Feature", "Absolute Loading"),
  align = "c") %>%
  group_rows("PC1", 1, 5) %>%
  group_rows("PC2", 6, 10) %>%
  group_rows("PC3", 11, 15) %>%
  kable_styling(latex_options=c("hold_position"), font_size=8)

# Scree plot
plot(pca_result$sdev, type='b',
  xlab='Principal Component',
  ylab='Standard Deviation',
  main='Scree Plot')

# PC scores
kable(pca_scores, digits = 2, caption = "Principal Component Scores by Drummer") %>%
  kable_styling(latex_options=c("hold_position"), font_size=8)

# Pairs plot
pairs(pca_result$x[, 1:3], main="PCA: First 3 Components")

# PC1 vs PC2 plot
ggplot(pca_scores, aes(x = PC1, y = PC2, label = drummer)) +
  geom_point(color = "steelblue", size = 3, alpha = 0.8) +
  geom_text(vjust = -0.7, size = 3, family = "serif") +
  theme_bw() +
  labs(
    title = "Drummers in PCA Space (PC1 vs PC2)",
    x = "PC1",
    y = "PC2"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),

```



```

    text = element_text(family = "serif")
  )

kable(pca_result$rotation[,1:3], digits = 3,
      caption = "Feature Loadings on First 3 PCs")

## KDE

# Calculating the bandwidth for velocity
bw_vel <- 1.06 * sd(all_hits$velocity) * length(all_hits$velocity)^(-1/5)

# Calculating the bandwidth for IOI
all_hits_ioi <- all_hits %>%
  filter(!is.na(inter_onset_interval), inter_onset_interval < 1)

bw_ioi <- 1.06 * sd(all_hits_ioi$inter_onset_interval) *
  length(all_hits_ioi$inter_onset_interval)^(-1/5)

# IOI KDE by drummer
ggplot(all_hits_ioi, aes(x = inter_onset_interval, color = drummer)) +
  geom_density(bw = bw_ioi) +
  theme_bw() +
  labs(title = "Inter-Onset Interval Distributions by Drummer (Gaussian KDE)",
       x = "Inter-Onset Interval (seconds)", y = "Density") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )

# Velocity KDE - faceted
ggplot(all_hits, aes(x = velocity)) +
  geom_density(fill = "steelblue", alpha = 0.6) +
  facet_wrap(~drummer, scales = "free_y", ncol = 5) +
  theme_bw() +
  labs(title = "Velocity Distributions by Drummer (Individual)",
       x = "Velocity", y = "Density") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )

# Extract KDE features
kde_summary <- all_hits %>%
  group_by(drummer) %>%
  summarise(
    # Velocity
    vel_mean = mean(velocity),

```

```

    vel_sd = sd(velocity),
    vel_range = max(velocity) - min(velocity),
    vel_cv = sd(velocity) / mean(velocity),
    # IOI
    ioi_mean = mean(inter_onset_interval, na.rm = TRUE),
    ioi_sd = sd(inter_onset_interval, na.rm = TRUE),
    ioi_cv = sd(inter_onset_interval, na.rm = TRUE) /
              mean(inter_onset_interval, na.rm = TRUE)
  )

# Merging with PCA
integration_df <- pca_scores %>%
  left_join(kde_summary, by = "drummer")

# Integration Table
kable(integration_df, digits = 2,
      caption = "Integration: PC Scores and Velocity/Timing Characteristics",
      ) %>%
  kable_styling(latex_options=c("hold_position"), font_size=8)

# Calculating correlations
cor_pc2_vel_mean <- cor(integration_df$PC2, integration_df$vel_mean, use = "complete.obs")
cor_pc2_vel_cv <- cor(integration_df$PC2, integration_df$vel_cv, use = "complete.obs")
cor_pc3_ioi_cv <- cor(integration_df$PC3, integration_df$ioi_cv, use = "complete.obs")
cor_pc1_vel_mean <- cor(integration_df$PC1, integration_df$vel_mean, use = "complete.obs")

#Correlation table
correlation_results <- data.frame(
  Relationship = c("PC2 vs Mean Velocity",
                  "PC2 vs Velocity CV",
                  "PC3 vs IOI CV",
                  "PC1 vs Mean Velocity"),
  Correlation = c(cor_pc2_vel_mean,
                  cor_pc2_vel_cv,
                  cor_pc3_ioi_cv,
                  cor_pc1_vel_mean),
  stringsAsFactors = FALSE
)

# Table
kable(correlation_results,
      digits = 3,
      caption = "Correlations Between PC Scores and Velocity/Timing Statistics",
      col.names = c("Relationship", "Correlation (r)"),
      align = c("l", "c"),
      ,
      row.names = FALSE) %>%

```

```

kable_styling(latex_options=c("hold_position"), font_size=8)

# Velocity KDE by drummer
ggplot(all_hits, aes(x = velocity, color = drummer)) +
  geom_density(bw = bw_vel) +
  theme_bw() +
  labs(title = "Velocity Distributions by Drummer (Gaussian KDE)",
       x = "Velocity", y = "Density") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )

# IOI KDE - faceted
all_hits_ioi %>%
  ggplot(aes(x = inter_onset_interval)) +
  geom_density(fill = "coral", alpha = 0.6) +
  facet_wrap(~drummer, scales = "free_y", ncol = 5) +
  theme_bw() +
  labs(title = "IOI Distributions by Drummer (Individual)",
       x = "Inter-Onset Interval (seconds)", y = "Density") +
  theme(
    plot.title = element_text(hjust = 0.5, family = "serif"),
    text = element_text(family = "serif"),
  )

## Hidden Markov Models

# Subsetting to one rock recording per drummer
rock_recordings <- all_hits %>%
  filter(grepl("rock", style, ignore.case = TRUE)) %>%
  group_by(drummer) %>%
  arrange(desc(n())) %>%
  slice(1) %>%
  distinct(drummer, id)

# Fitting a 3-state HMM for each drummer
# the 3 states are soft, medium, loud dynamics
hmm_results <- list()

for (i in 1:nrow(rock_recordings)) {
  d <- rock_recordings$drummer[i]
  rec_id <- rock_recordings$id[i]

  rec_data <- all_hits %>%
    filter(id == rec_id) %>%
    arrange(time) %>%

```

```

dplyr::select(velocity)

if (nrow(rec_data) > 100) {
  set.seed(123)
  hmm_model <- depmix(velocity ~ 1,
                     data = rec_data,
                     nstates = 3,
                     family = gaussian())

  hmm_fit <- tryCatch(
    fit(hmm_model, verbose = FALSE),
    error = function(e) { return(NULL) }
  )

  if (!is.null(hmm_fit)) {
    # Extracting parameters
    trans_matrix <- matrix(NA, nrow = 3, ncol = 3)
    for (j in 1:3) {
      trans_matrix[j, ] <- hmm_fit@transition[[j]]@parameters$coefficients
    }

    state_means <- numeric(3)
    for (j in 1:3) {
      state_means[j] <- hmm_fit@response[[j]][[1]]@parameters$coefficients[1]
    }

    state_order <- order(state_means)
    state_means_sorted <- state_means[state_order]
    trans_sorted <- trans_matrix[state_order, state_order]

    # Getting our results here to display
    hmm_results[[as.character(d)]] <- list(
      drummer = as.character(d),
      n_hits = nrow(rec_data),
      state_means = state_means_sorted,
      transition_matrix = trans_sorted,
      persistence = mean(diag(trans_sorted)),
      switching = mean(trans_sorted[row(trans_sorted) != col(trans_sorted)])
    )
  }
}

# summary table (Not displaying this, putting it into a kable)
hmm_summary <- data.frame(
  drummer = sapply(hmm_results, function(x) x$drummer),
  n_hits = sapply(hmm_results, function(x) x$n_hits),

```

```

soft_state = apply(hmm_results, function(x) round(x$state_means[1], 1)),
medium_state = apply(hmm_results, function(x) round(x$state_means[2], 1)),
loud_state = apply(hmm_results, function(x) round(x$state_means[3], 1)),
persistence = apply(hmm_results, function(x) round(x$persistence, 3)),
switching = apply(hmm_results, function(x) round(x$switching, 3))
)

# HMM Summary Table
kable(hmm_summary,
      caption = "HMM Characteristics: Rock Performances",
      col.names = c("Drummer", "Hits", "Soft", "Medium", "Loud",
                    "Persistence", "Switching"),
      digits = 2) %>%
kable_styling(latex_options=c("hold_position"), font_size=8)

# Merging the HMM results with PCA/KDE integration
integration_hmm <- integration_df %>%
  left_join(hmm_summary, by = "drummer")

kable(integration_hmm %>%
      dplyr::select(drummer, PC2, vel_mean, vel_cv, loud_state, persistence),
      digits = 2,
      caption = "HMM Integration with PCA and KDE Results") %>%
kable_styling(latex_options=c("hold_position"), font_size=8)

#Using the same colors for consistency
steelblue_pal <- colorRampPalette(c("white", "steelblue"))(20)

# Plotting a single transition matrix for drummer1
par(mar = c(4, 4, 3, 2), family = "serif")
if ("drummer1" %in% names(hmm_results)) {
  A <- hmm_results[["drummer1"]]$transition_matrix

  # Heatmap
  image(1:3, 1:3, t(A)[,3:1], col = steelblue_pal,
        main = "Transition Matrix: Drummer 1",
        xlab = "To State",
        ylab = "From State", axes = FALSE)
  axis(1, at = 1:3, labels = c("Soft", "Med", "Loud"), cex.axis = 1)
  axis(2, at = 1:3, labels = c("Loud", "Med", "Soft"), cex.axis = 1)

  # Probability values
  for (i in 1:3) {
    for (j in 1:3) {
      text(j, 4-i, sprintf("%.2f", A[i,j]), col = "black", cex = 1.2, font = 2)
    }
  }
}

```

```

# Caption
mtext("Diagonal values = state persistence | Off-diagonal = switching probability",
      side = 1, line = 4, cex = 0.9)
}

## Appendix C Visualizations

kable(all_hits %>%
      count(drummer) %>%
      arrange(desc(n)),
      caption = "Number of Hits per Drummer")

recordings_per_drummer <- all_hits %>%
  group_by(drummer) %>%
  summarise(num_recordings = n_distinct(id)) %>%
  arrange(desc(num_recordings))

kable(recordings_per_drummer,
      caption = "Number of Recordings per Drummer",
      col.names = c("Drummer", "Recordings"))

#turning the steel blue color to a gradient (one-sided)
steelblue_pal <- colorRampPalette(c("white", "steelblue"))(20)

# Plotting the transition matrices
par(mfrow = c(3, 4),
    mar = c(3, 3, 2, 1),
    family = "serif") # match your ggplot serif theme

for (d in names(hmm_results)) {
  A <- hmm_results[[d]]$transition_matrix

  # Heatmap
  image(1:3, 1:3, t(A)[,3:1],
        col = steelblue_pal,
        main = d,
        xlab = "", ylab = "",
        axes = FALSE)

  axis(1, at = 1:3, labels = c("Soft", "Med", "Loud"),
        cex.axis = 0.8)
  axis(2, at = 1:3, labels = c("Loud", "Med", "Soft"),
        cex.axis = 0.8)

  # Probability values
  for (i in 1:3) {
    for (j in 1:3) {

```

```

        text(j, 4-i, sprintf("%.2f", A[i,j]),
              col = "black", cex = 0.9)
    }
}
}

#Legend
plot.new()
text(0.5, 0.75, "Transition Probabilities", cex = 1.3, font = 2)
text(0.5, 0.55, "Rows = From State", cex = 1)
text(0.5, 0.45, "Columns = To State", cex = 1)
text(0.5, 0.35, "Diagonal = Persistence", cex = 1)
text(0.5, 0.25, "Off-diagonal = Switching", cex = 1)

```